Ema Farshi, CS and Data Science

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EDUCATION

University of Toronto | Honours Bachelor of Science Expected 2029

Computer Science and Data Science | GPA: 4.0/4.0
Relevant Coursework: Foundations of Computer Science I and II, Calculus with Proofs, Statistical Reasoning and Data Science
In Progress: Software Design, Theory of Computation, Probability and Statistics I, Multivariable Calculus with Proofs

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, R
Frameworks & Libraries: React, FastAPI, scikit-learn, Playwright, BeautifulSoup, PyGame, tkinter, pytest
Tools & Technologies: Git, GitHub, PostgreSQL, Node.js, Steel.dev
Methodologies: Spec-Driven Development, Agile Development
Concepts: Data Structures and Algorithms, Object-Oriented Programming, Machine Learning, Data Analysis

EXPERIENCE

Teaching Assistant - Linear Algebra | University of Toronto Sept. 2026 - Dec. 2026

- Support students in developing their understanding of linear algebra through in-class instruction and online discussion-board assistance
- Explain mathematical concepts and problem-solving strategies to students with varying levels of experience
- Grade weekly coursework and provide feedback while assisting during lectures

PROJECTS

ZoomIn | React, FastAPI, Python, scikit-learn, Playwright, Steel.dev Sept. 2026

- Directed development of a full-stack subscription management prototype at the Battle of the Schools AI/ML Hackathon, collaborating with teammates responsible for UI/UX design
- Guided development of Playwright and Steel.dev browser automation for assisted subscription cancellations, with pauses for authentication and final user confirmation
- Incorporated website-scanning functionality using TF-IDF/logistic regression, BeautifulSoup DOM heuristics, and Google Safe Browsing to identify potentially deceptive payment practices

Pink Desktop Calculator | Python, tkinter, pytest

- Built a full-expression calculator supporting operator precedence, powers, percentages, degree-based trigonometry, and implicit multiplication without eval
- Applied spec-driven development to translate functional requirements into software architecture, implementation tasks, and automated testing

Recipe Recommendation System | Python, PyGame

- Collaborated on an interactive recipe recommendation application using graph- and tree-based data structures to organize information and generate recommendations from user preferences
- Implemented search and filtering across ingredients, dietary restrictions, excluded ingredients, and recipe categories, with a PyGame interface

LEADERSHIP & INVOLVEMENT

Special Events Associate - Computer Science Students' Union (CSSU) May 2026 - Present

- Help organize major events for computer science students, including orientation, semi-formal events, and an undergraduate research conference

Seminars Executive - Undergraduate Research Students' Association (URSA) June 2026 - Present

- Help plan research seminars and major events, including a 300+ attendee undergraduate research session and the Tri-Campus Undergraduate Research Conference